

ANKITA SETHI

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PROFESSIONAL EXPERIENCE

- Software Engineer** (Founding ML Engineer) | *Aurora IQ Inc. (Early-stage AI startup)* | Remote (U.S) Feb 2026 – Present
- Built and deployed an **end-to-end ML and computer vision pipeline** for an AI-driven product discovery platform, processing **20K+ images** from ingestion to recommendation using FastAPI and SQLAlchemy.
 - Trained multimodal embedding models** and built **vector search** retrieval, **evaluating against baseline rankers via offline A/B tests** with significance testing and lifting recommendation relevance by ~25%.
 - Designed feature and data pipelines on **PostgreSQL**, **AWS S3** and **vector DB**, reducing inference latency by ~40% through query optimization and indexing.
- Data Scientist** (Research Assistant) | *Stony Brook University* | *Stony Brook, NY (Remote)* Jul 2025 – Apr 2026
- Engineered an end-to-end **ML pipeline and evaluation framework** for content segmentation using Python, Django and Redis, reducing response latency by 35% and improving stability across diverse pages.
 - Monitored model performance and ran error analysis** on segmentation outputs, identifying failure patterns and improving robustness through prompt and pipeline iteration.
 - Integrated **Gemini 2.5** and **LangChain** with prompt versioning and experiment tracking, improving segmentation accuracy from 72% to 85% on a held-out dataset across multiple model versions.
- Data Scientist** (Graduate Research Assistant) | *Stony Brook University* | *Stony Brook, NY* Aug 2024 – May 2025
- Led **LLM and ML model development** for accessibility, evaluating performance across **100+ test cases using hypothesis testing to compare model variants** across documents and complex web content.
 - Built monitoring metrics for latency, failure categories, and accuracy drift**, with delivery via Chrome extensions and NVDA plugins backed by Django pipelines.
- Data Engineer** | *Accenture Private Limited* | *Bangalore, India* Jun 2021 – Aug 2023
- Built and automated structured data extraction by deploying BERT-based NLP models** across 2M+ records.
 - Performed **data analysis and feature engineering** to improve **model accuracy and extraction quality**.
 - Engineered and maintained large scale **ETL** pipelines using **Talend** and **Snowflake**, increasing data throughput by 35% and reducing production pipeline latency.
 - Built 15+ **Tableau** dashboards to monitor **ML KPIs**, pipeline health, data/model anomalies, cutting issue identification time by 40%.

PROJECTS AND RESEARCH

- Member Aware Question Answering System** Nov 2025
- Built a retrieval-augmented QA system using Gemini 2.5, Django and JSON for more than 3.5K member messages.
 - Implemented context ranking, update detection, and prompt construction with median latency under 2s, **ran offline evaluations comparing retrieval and prompt variants**.
 - Structured scalable pipeline for message extraction, feature preparation, deployed the service on Render with ongoing latency tracking.
- Multi-Agent Approach for Detecting Hallucination in LLMs** Sep 2024
- Designed a multi-agent framework for **LLM hallucination detection** using concept inference and NER, **improving detection F1 by 18%** across evaluation benchmarks.
 - Scaled domain coverage from 7 to 57 via self-familiarity scoring, increasing detection consistency across varied text types by 22%.
- Smart Energy Assistant for Smart Homes** Feb 2024
- Built an ARIMA time-series forecasting model for appliance usage from weather and historical data, **tested via walk-forward cross-validation** across 1000+ observations, reducing **MAE by 28%** vs. a naive seasonal baseline.
 - Layered GPT-4 anomaly detection on top of forecasts, achieving **$R^2 = 0.737$** and outperforming ARIMA-only and naive baselines on held-out data.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++
NLP & LLMs: RAG, LangChain, Agentic Pipelines, Prompt Engineering, LLM Evaluation, NLP, Embeddings, Vector Search
ML & Deep Learning: PyTorch, HuggingFace, scikit-learn, NumPy, Pandas, Transformers, Computer Vision, OpenCV
Statistics & Experimentation: A/B Testing, Experimental Design, Regression, Classification, Optimization, Time-Series Forecasting, Hypothesis Testing
Data & Cloud: Snowflake, PostgreSQL, Redis, Spark, Airflow, ETL Pipelines, Data Modeling, AWS (S3), Databricks
BI & Visualization: Looker, Tableau, Dashboarding, KPI Monitoring
Backend & DevOps: FastAPI, Django, REST APIs, Docker, CI/CD, Jenkins, Git

EDUCATION

Stony Brook University | **Stony Brook, NY** | *Master of Science in Computer Science* Aug 2023 - May 2025
GITAM University | **Visakhapatnam, India** | *Bachelor of Technology in Computer Science and Engineering* Jun 2017 - Jun 2021